



Clickwright

Three AI agents turn a feature spec into live ClickHouse tables, documented context, and PM-ready insights — with every decision traced and human-approved.

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Team Shard Spartans

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ClickHouse Cloud

Claude (Anthropic)

Langfuse

TypeScript

THE PROBLEM

Getting from raw events to trustworthy analytics is slow, brittle, and manual.

Schema design is guesswork

Engineers pick ordering keys and types by intuition. Wrong choices are immutable — a bad key means rebuilding the table.

Context lives in people's heads

Metric definitions, data quirks, known issues — scattered across Slack, docs, and tribal knowledge. Every new analyst starts from zero.

AI analytics hallucinate numbers

LLM-generated insights look plausible but cite figures that aren't in the data. A PM acts on a number that doesn't exist.

No audit trail

When an insight is wrong, there's no way to trace it back to the query, the schema decision, or the context that shaped it.

THE SOLUTION

Spec in. Insight out. Every step traced.

An agentic pipeline that instruments, documents, and analyses — with human gates at every irreversible decision.



Every number traces back to a SQL result. Every schema decision traces to a Langfuse span. Every context change is versioned and attributed. Nothing is a black box.

Three agents. One shared brain.

Instrumentation

01

- Profiles every field (types, nulls, cardinality)
- Deterministic baseline DDL from stats
- LLM optimizes via architecture review skill
- Codecs, ordering keys, cross-table coherence
- EXPLAIN AST dry-run before execution
- DML retry separate from schema retry

Context

02

- Append-only versioned knowledge store
- Table docs synthesized from measurements
- LLM writes spec summaries + metric defs
- Detects contradictions with existing context
- Every future agent inherits what was learned
- Semantic retrieval for mid-analysis lookups

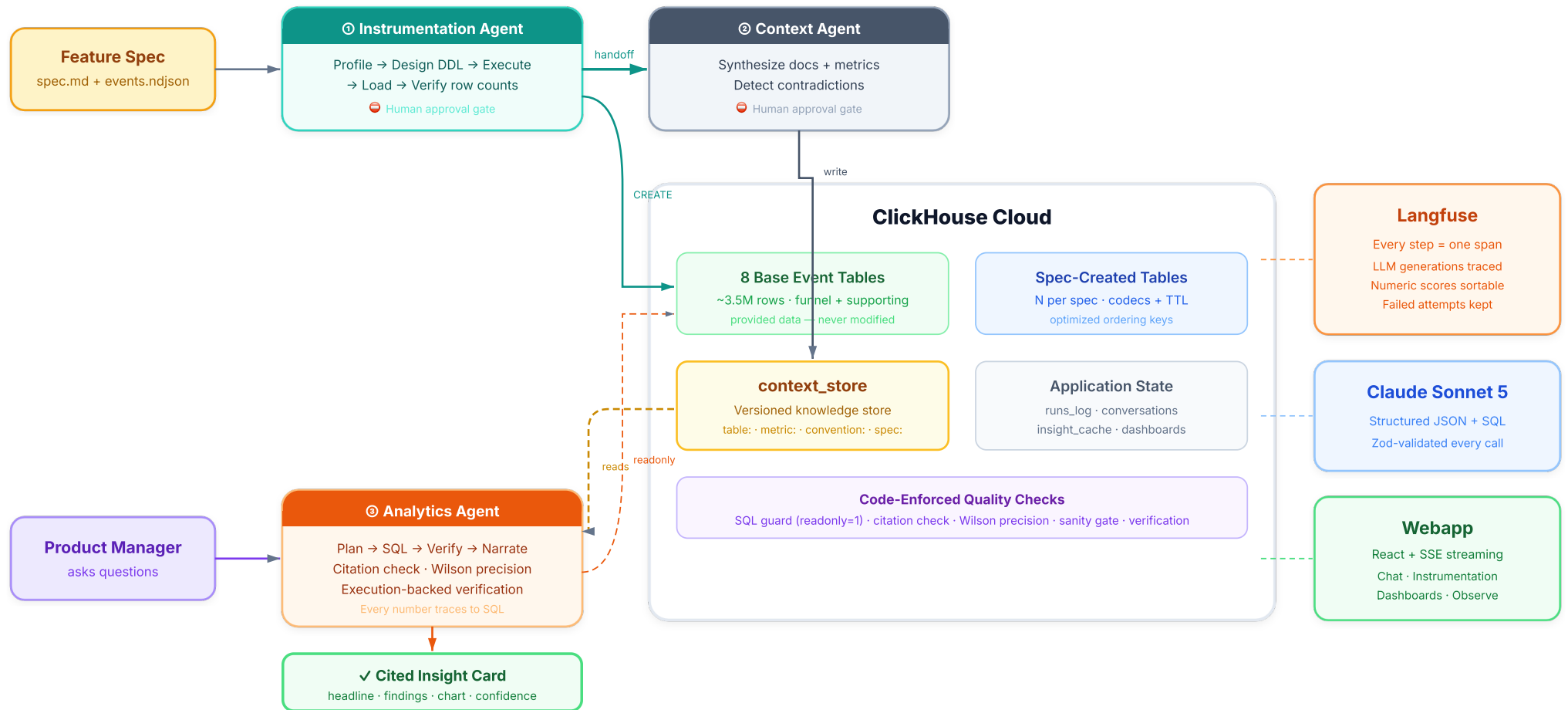
Analytics

03

- Plans ≤ 4 tasks with proactive segmentation
- Concurrent SQL with self-healing retries
- Full-set digests computed in ClickHouse
- Citation check: every number verified
- Independent query cross-checks headline
- Wilson confidence intervals on every rate

Agents share state through **context_store** in ClickHouse — never call each other directly. Runs are serialized; tasks within a run are concurrent.

How the three agents collaborate



Seven layers between the LLM and the PM.

Quality is the hard constraint. A slower right answer beats a fast wrong one.



SQL Guard

readonly=1, banned keywords, single statement, LIMIT cap. The model cannot mutate data.



Full-Set Digest

Population stats computed over every row in ClickHouse. No extrapolation from samples.



Sanity Gate

Rates >100% flagged. Empty results dropped. Low sample sizes warned. All over the full set.



Citation Check

Every number in prose must exist in the SQL results — or be a verified arithmetic of two that do.



Wilson Precision

95% confidence intervals computed from the actual denominator. Reported with the figure, not separately.



Execution-Backed Verification

An independently written query recomputes the headline figure. Disagreement drops confidence to low.



Established Figures

Follow-up answers carry prior figures + denominators. A changed denominator is explained, never silent.



Human Approval Gates

DDL and context changes require explicit human approval. The agent proposes; the human decides.

Built different, by design.



ClickHouse Best-Practice DDL

Schemas are **measured, not guessed**. Code profiles every field — types, null rates, cardinality, ranges — and builds a deterministic baseline with correct LowCardinality, Decimal64 for money, and data-driven ordering keys. The LLM optimizes codecs, partitioning, and cross-table coherence in one call — guided by the **ClickHouse architecture review skill**. Every DDL is dry-run through EXPLAIN AST before execution. If the LLM fails, the measured baseline ships unchanged.



Cross-Conversation Context

Follow-ups carry **prior SQL, established figures, their denominators, and source tables** from earlier turns. This prevents silent denominator drift — if one answer reports UAE conversion as 56.6%, the next turn knows both the number and the *n* it rests on. A changed denominator is **explained, never silent**. Failed tasks are also forwarded so the planner doesn't repeat impossible work.



Wilson Score Validation

Every rate is **classified** (proportion, mean, quantile, ratio) — only proportions get a 95% Wilson interval from the actual denominator. Confidence is **computed, never asked of the model**: widest interval, sanity flags, citation retries, and independent verification query agreement are combined into a deterministic score. $\pm 10\text{pp+}$ or a verification disagreement drops confidence to low.



These are not configuration options — they are **code-enforced by default**. Every DDL, every insight, and every follow-up answer passes through them automatically.

BUILT WITH

Production-grade from day one.

ClickHouse Cloud

DATABASE

Event tables, knowledge store, run history, insight cache, dashboards — all in one engine. Codecs, TTL, and ordering keys optimized per table.

Claude Sonnet 5

LLM

Best structured-output reliability for DDL + SQL + JSON. Effort pinned to medium — schema-validated prompts don't need extended thinking.

Langfuse Cloud

TRACING

Every span, generation, and score queryable. Numeric scores (self-heal attempts, citation failures) as sortable columns.

TypeScript

BACKEND

Strong typing + Zod runtime validation on every LLM output. Async-native for concurrent task execution.

React + Vite

FRONTEND

SSE streaming for live pipeline steps. Chat UI with insight cards, charts, segment tables, and confidence badges.

Zod

VALIDATION

Every LLM response is parsed through a strict schema. Malformed output triggers self-healing retry with the verbatim error.

~30s

Spec → live tables

~15s

Question → cited insight

~0.6s

Cached repeat question

100%

Numbers traceable to SQL

Not dashboards. Answers.

Headline with the number that matters

Every insight leads with the one figure the PM needs — not a data dump.

"Express Checkout lifts checkout-to-purchase conversion by 12.3pp over standard (67.0% vs 54.7%, n=1,007 vs 3,412)."

Findings tagged by type

Drivers (what explains it), segments (where it concentrates), caveats (what to watch), known issues (linked by ID).

SEGMENT: "The gap is widest on iOS in the UAE — 67.7% Express vs 41.8% standard — consistent with K1 (OTP autofill regression)."

Charts and tables from the SQL

Bar/line charts and segment tables rendered directly from query results. Value formats (% , count, currency) inferred from column names and SQL.

Chart: "Step-through rate by funnel stage: iOS vs Android" — 8 bars, lowest highlighted in coral.

Confidence you can trust

Computed from Wilson intervals, sanity flags, citation retries, and an independent verification query. Never the model's self-assessment.

Confidence: HIGH — "±2.1pp on conversion_rate (n=6,715); an independently written query reproduced the figure."

Spec in. **Insight out.** Every step traced.

Clickwright makes analytics trustworthy by construction — not by hoping the LLM gets it right, but by making it prove that it did.

REPO

github.com/bjpadhy/Clickwright

TRACES

Every run linked to Langfuse

STACK

ClickHouse · Claude · Langfuse · TypeScript

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